MAJOR PROJECT REPORT

Smart Fake News Detection System

A Machine Learning Based Approach for Classifying News Articles

Project Domain: Artificial Intelligence and Machine Learning

1. Abstract

The Smart Fake News Detection System is a machine learning project designed to classify news articles as real or fake based on their text. The system uses text preprocessing, TF-IDF feature extraction and classification algorithms. Logistic Regression and Multinomial Naive Bayes were trained and evaluated. Logistic Regression achieved the highest F1-score and was selected for the application.

2. Introduction

The rapid growth of online news and social media has made it easier for misleading information to spread. Fake news can create confusion and affect public understanding. This project demonstrates how machine learning can help identify patterns in news text and classify articles into real and fake categories.

3. Objectives

• Prepare and clean a labelled news dataset.
• Convert news text into numerical features using TF-IDF.
• Train Logistic Regression and Multinomial Naive Bayes models.
• Compare model performance using evaluation metrics.
• Build a simple web application for text classification.

4. Dataset

The project uses the ISOT Fake News Dataset, containing separate real and fake news article files. The two files were combined into one dataset. Fake news was assigned label 0 and real news was assigned label 1. The dataset is historical and does not represent all current news sources or events.

5. Data Preprocessing

The dataset was checked for missing and duplicate records. News text was cleaned by converting it to lowercase, removing web links and removing non-alphabetic characters. Headline and article text were combined to prepare the input for model training.

6. Feature Extraction

TF-IDF (Term Frequency-Inverse Document Frequency) was used to convert the cleaned news text into numerical features. The vectorizer was configured with a maximum of 5,000 features and English stop-word removal.

7. Model Training

The dataset was divided into training and testing sets using an 80:20 split with stratification. Two machine learning algorithms were trained: Logistic Regression and Multinomial Naive Bayes. Both models were evaluated on the same test data.

8. Model Evaluation

Metric	Logistic Regression	Multinomial Naive Bayes
Accuracy	98.43%	93.04%

Precision	97.86%	93.76%
Recall	99.32%	93.54%
F1-score	98.58%	93.65%

9. Result and Model Selection

Logistic Regression achieved an accuracy of 98.43% and an F1-score of 98.58%. Multinomial Naive Bayes achieved an accuracy of 93.04% and an F1-score of 93.65%. Logistic Regression performed better across the reported evaluation metrics and was selected as the final model for the application.

10. Application Development

A Streamlit web application was developed. Users can enter a news article or text, and the application uses the saved TF-IDF vectorizer and Logistic Regression model to generate a predicted category. The application also displays a model confidence estimate.

11. Tools and Technologies

Python, Google Colab, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Joblib, Streamlit and GitHub.

12. Limitations

The model learns patterns from its training dataset and may not generalize to every topic, writing style or newly emerging event. Its prediction is not proof that a news article is true or false. The confidence score is a model estimate, not a verified probability that the article is factual.

13. Future Scope

The system could be improved by using larger and more recent datasets, adding more advanced language models, supporting multiple languages and integrating reliable external fact-checking sources.

14. Conclusion

The project demonstrates a complete machine learning workflow for news text classification, from data preparation and feature extraction to model evaluation and application development. Logistic Regression was selected because it achieved the best results among the two evaluated models. The application is intended as an educational classification tool and should not replace independent fact-checking.